

# Failure 3 — the cost wall

*Better rerankers exist. You can't afford to run them.*

## LATENCY TAX

40x

slower per query

*cross-encoder → LLM reranker*

Production search has a second-scale latency budget, not a minute-scale one.

*You can't afford the LLM reranker on 10–20 candidates. Not on 1,000.*

**Better rerankers don't fix bad retrievers — they make the gap more expensive to ignore.**

## TIME TO RANK 1,000 DOCUMENTS

cross-encoder reranker

~3 s

**LLM reranker (7B param)**



1–2 s

*production budget*